CSCI 270 — Applied Artificial Intelligence

Fall 2026

Units	3
Instructor Name	Anurag Tiwari
Office	Harbor Walk East Rm. 304
Office Hours	MW 11:30 am -12:20 pm (Scheduled) Vitual : Teams
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Lecture Section

Details

Section	Room	Time
Section 1 (CRN-11578)	HWEA 300	MWF 12:30 pm – 1:20 pm

Course Description

This course provides students with a practical foundation in artificial intelligence, emphasizing the application of computational methods to real world problems. The course introduces the relationship between artificial intelligence and human cognition and provides experience with data driven problem solving, machine learning, generative AI, optimization, and applied artificial intelligence systems.

Overview

- Examine how artificial intelligence attempts to model human thought processes and consider the potential and limitations of these approaches.
- Explore data classification, regression analysis, clustering, and time series forecasting.
- Work with supervised and unsupervised learning and examine methods for training and evaluating models.
- Apply data scaling, normalization, and distance metrics to multidimensional data.
- Progress from foundational machine learning methods such as K Means clustering and linear regression to radial basis function networks.
- Explore optimization through simulated annealing and the Nelder Mead algorithm.

- Investigate discrete optimization problems, including the Traveling Salesman and Knapsack problems.
- Apply artificial intelligence to financial data analysis, image recognition, optical character recognition, genetic algorithms, and swarm intelligence through applications such as the Boids flocking model.
- Explore foundations of Generative AI in text and images.
- Complete coding assignments and projects involving implementation, analysis, evaluation, and optimization.
- Develop a portfolio demonstrating practical artificial intelligence development and problem solving.

Course Prerequisites

CSCI 220 & 220L *or* CSCI 218 & 218L

Course Objectives

The course is designed to enable students to:

1. Establish a working foundation in artificial intelligence terminology, concepts, and computational approaches.
2. Develop proficiency in using data and computational representations to formulate artificial intelligence problems.
3. Introduce systematic approaches for developing, training, testing, and refining artificial intelligence models.
4. Develop practical programming skills for implementing artificial intelligence and machine learning methods.
5. Provide experience with mathematical and computational techniques that support artificial intelligence, including normalization and distance measurement.
6. Introduce model optimization strategies for improving solutions to continuous and discrete computational problems.
7. Connect artificial intelligence methods to applications in areas such as finance, computer vision, text recognition, logistics, and resource management.
8. Understand the design and logic of generative AI through text and images.
9. Develop the ability to investigate artificial intelligence solutions through experimentation, analysis, and evaluation.
10. Provide experience designing and completing substantial artificial intelligence projects that integrate multiple course concepts.
11. Prepare students to independently apply artificial intelligence knowledge to unfamiliar problems and emerging technologies.

Student Learning Outcomes

Upon successful completion of the course, students will be able to:

1. **Analyze** real world problems to determine relevant artificial intelligence requirements and computational approaches.
2. **Differentiate** between supervised and unsupervised learning approaches and **justify** their selection for appropriate applications.
3. **Implement** classification, regression, clustering, and time series methods for appropriate data driven problems.
4. **Calculate** and **interpret** normalization results and distance measurements within multidimensional data spaces.
5. **Implement** K Means clustering, linear regression, and radial basis function networks using appropriate data and computational procedures.
6. **Evaluate** artificial intelligence models using appropriate performance measures and **interpret** the implications of model results.
7. **Optimize** computational solutions using simulated annealing and the Nelder Mead algorithm and **compare** the effectiveness of different optimization approaches.

8. **Solve** discrete optimization problems such as the Traveling Salesman and Knapsack problems using appropriate computational strategies.
9. **Apply** genetic algorithms and swarm intelligence techniques to computational problems and **evaluate** the resulting solutions.
10. **Understand** the foundations of Generative AI through tokenization of text and vectorization and slicing of image data.
11. **Develop** artificial intelligence applications involving financial data, image recognition, optical character recognition, and other real world datasets.
12. **Analyze** the behavior and performance of artificial intelligence systems, including applications based on the Boids flocking model.
13. **Integrate** multiple artificial intelligence techniques to develop solutions for complex practical problems.
14. **Assess** the strengths, limitations, accuracy, and applicability of artificial intelligence solutions within specific problem domains.
15. **Design** and **implement** artificial intelligence projects that demonstrate appropriate selection and integration of computational methods.
16. **Create** a portfolio of completed artificial intelligence projects that demonstrates technical proficiency, problem solving, experimentation, and practical application.

Required Texts, Equipment, Technology

Details

Resource	Details
Official Reading	• Artificial Intelligence for Humans, Vol 1: Fundamental Algorithms Free ebook. Book codebase. • Lecture Slides
Recommended Reading	Turing Paper
Practice	GeeksForGeeks, W3Resources

Exams

Exams are cumulative from the 1st day of class. Both the midterm and final exams are in-person handwritten exams. There can be (but not limited to) code writing, code alterations and debugging, MCQs, Fill-in-the-Blanks and True-False questions. Exams are Closed book with no access allowed to cellphones, laptops, smartwatches or cheat-sheets.

Details

Exam	Room	Section 1 (CRN-11151)
Mid-Term	HWEA 300	12:30 pm – 1:20 pm Oct 2nd (Friday) *Subject to Change*
Finals	HWEA 300	1:00 pm – 3:00 pm Dec 6th (Sunday)

Methods of Instruction

This course is delivered through in-person lectures with in-class coding tasks and demonstrations. Students are expected to complete assigned readings prior to each class and practice programming outside of class hours.

Course Policies

Extra Credit

Students may complete the following Coursera courses to receive extra credit in exams as per the details below:

- Python For DataScience, AI & Developer - 50% of Midterms
- Machine Learning with Python - 30% of Finals

Student who have completed these courses prior to this semester and would like to apply for extra credit may discuss alternate courses for extra credit, with the Instructor. Only courses completed during the current semester, may be redeemed for extra credit.

Intellectual Property Policy

All in-class course content including exams are the intellectual property of the instructor. Screenshots, notes, and copies may be shared ONLY with currently enrolled students and MUST be deleted at the end of the semester or stored in private repositories for personal use. Completed projects may be stored in public repositories at the student's discresion.

Discovery of course content on online platforms, accessible to the public, even after course completion (even after graduation) will be reported to COFC officials as academic dishonesty.

Class Decorum and Behavior

Students are requested to maintain a respectful and productive environment. Use of electronic devices should be for class-related activities only. Be on time and be prepared for class/lab.

Disruptive behavior will result in expulsion from the class at the discretion of the instructor/teaching assistant. Repeat offenders will be recommended for disciplinary action.

Course Assignments

Students are expected to submit all assignments and projects by the deadline provided by the instructor. In case of an unforeseen emergency, the instructor may grant an exception. Contact the instructor in advance if at all possible.

> All assignments and projects may go through **code review sessions** after their submission deadlines. Inability to explain your code is an indication that the work was not yours.

Late work: Late submissions for assignments will receive a 25% penalty per day for up to 3 days. After 3 days late submissions are not accepted. Late lab submissions are not accepted.

Grading Policy

Details

Component	Weight
Educational Research-Study Activities	2%
Attendance	5%

Assignments	18%
Projects	20%
Midterm Exam	25%
Final Exam	30%

Grade Scale

Details

Range	Grade
≥ 93.00	A
90.00 – 92.99	A-
87.00 – 89.99	B+
83.00 – 86.99	B
80.00 – 82.99	B-
77.00 – 79.99	C+
73.00 – 76.99	C
70.00 – 72.99	C-
67.00 – 69.99	D+
63.00 – 66.99	D
60.00 – 62.99	D-
≤ 59.99	F

Attendance Policy

Students are expected to attend class regularly. Excessive absences may result in a W (failure by withdrawal) or a WA grade on the recommendation of the instructor to the school officials. Attending class will contribute towards the final grade directly.

Exceptions will only be made for emergencies.

Scheduling non-emergent appointment will not be treated as a valid excuse for missed attendance.

Academic Integrity Statement

> As members of the College of Charleston community, we affirm, embrace and hold ourselves accountable to the core values of integrity, academic excellence, liberal arts education, respect for the individual student, student centeredness, innovation, and public mission. Academic integrity is the academic community's commitment to honesty, fairness, respect, responsibility, courage, trustworthiness, and ethical conduct which serves as a guiding principle for all faculty,

> staff, and students in the human pursuit of truth and knowledge. Our academic work reflects not only what we have learned but also how we have engaged with the material.
>
> The College of Charleston expects that every student and community member has a responsibility to uphold the standards of the Honor Code, as outlined in the Student Handbook. In pursuit of academic learning, you are expected to reference the work of other scholars, and complete your own academic work, while utilizing appropriate resources for assistance as outlined by each course instructor. This collaborative spirit enhances our learning environment and fosters a culture of trust. Any concerns regarding academic honesty will be reported to the Office of the Dean of Students and addressed through the conduct process. Your commitment to these practices and expectations is essential in cultivating a vibrant campus culture that balances trust with the pursuit of knowledge, ultimately contributing to a strong foundation of academic excellence at the College of Charleston. If you have any questions about these expectations, please reach out to your instructor for clarification.

Instructor Policy on Academic Integrity

> **Generative Artificial Intelligence (GAI) tools are FULLY permitted for any deliverable work in this class.** Students are encouraged to use GenAI tools for deliverables in this course. However, deliverables are the responsibility of student and not the tools they use. Inability to explain all portions of the code upon detection of abuse, will result in a zero grade on the deliverable.

Students are bound by honor and by their enrollment at the College to abide by the specific policies of the Honor Code and the Code of Conduct, and to report any violations of the Honor System. All work submitted in this course must be your own or your assigned teams, completed in accordance with the College's Honor Code. You may not engage in unauthorized collaborations.

Students found responsible for academic dishonesty by the College of Charleston Honor Board will receive an X-F grade — indicating failure in the course due to academic dishonesty. This is recorded and will remain on the student's transcripts for 2 years, after which the student may request removal of the "X" resulting in an "F" grade. Other consequences include academic probation, temporary suspension, or permanent dismissal.

Note from Instructor

> **External help from friends, seniors, etc, that are strictly not part of your assigned team, and copy pasting code from websites and tools is still considered plagiarism.**

- Do not copy paste the homework/assignment/project prompts into an LLM and submit the regurgitated answer/code from the LLM.
- Use the assignment prompts to step by step break down the problem and “talk” with your LLM of choice. Understand the “WHY” at every step.
- Each assignment will require a PDF report to be submitted with it detailing which LLM you chose (if you had to) and why you and the LLM made the choices that you did.
  - For Example - what model was used to make a certain prediction? Linear Regression? Why? Were there outliers in the DATA? Were they removed from the training data?
- THIS BEING SAID, I would still prefer it if you tried to solve the assignments without help.

AI chatbot Classroom Activity and Educational Research Notice

Activity Overview

As a required component of this curriculum, students will explore certain topics by interacting with a custom-designed AI powered chatbot, before or after receiving instructions on the same topic by the Instructor. Following the interactions, students will complete a topic focused quiz and feedback survey. This activity is designed to enhance the students understanding of the course concepts, and develop practical AI literacy.

Educational Research Notice

Data generated from this course activity will be used in an educational research study evaluating different AI-chatbots design frameworks and how they impact student learning.

This activity is a required portion of the course, to be treated just like an assignment or deliverable. However, to protect student privacy, adhering to the College of Charleston's strict ethical standards, the following protections are in place:

- Complete De-identification: All student identifiers will be stripped from the collected data prior to any research analysis - making the research dataset completely anonymous.
- No Impact on Student Grades: Student performance on the quiz and answers on the survey will be strictly used to measure chatbot effectiveness and impact. They will have no impact on the the student grade in this course or academic standing.
- Strict Adherence to CofC and IRB Guidelines: This activity and the associated educational research will adhere strictly to the guidelines, standards and protocols set in place by the College of Charleston's Institutional Review Board and the Office of Research Integrity (ORI)

Student will be awarded 2% credit towards their final grade upon completion of all the Classroom Activity sessions.

Please contact the instructor for any questions regarding this activity or associated research.

Accommodations for Students with Disabilities

The College of Charleston is committed to providing access and inclusion for all persons with disabilities. Students with disabilities who anticipate or experience disability-related barriers in the academic setting should notify their professor, and initiate contact with the Center for Disability Services (CDS) to move forward with the college's established accommodation process (email: cds@cofc.edu; phone: 843-953-1431; location: Lightsey Center 104).

Students already connected with CDS for academic-related accommodations should share their Professor Notification Letter (PNL) with each professor as early in the semester as possible so that accommodations can be made promptly. The PNL informs faculty of potential course access and accommodations that might be necessary and reasonable. It is the student's responsibility to follow this process each semester.

OAKS (3.10)

OAKS, including Gradebook, will be used for this course throughout the semester to provide the syllabus and class materials and grades for each assignment, which will be regularly posted.

Digital Accessibility

> The College's Digital Accessibility Initiative (DAI) reflects our commitment not only to meeting federal requirements but also to fostering a more inclusive and supportive learning environment for everyone.
>
> Consistent with access for all, this course uses Ally in OAKS to ensure digital content is as accessible as possible. Captions or transcripts are provided on all videos and audio recordings in the course. If you find online course materials are not meeting your needs, please reach out to me so we can remediate the content to ensure accessibility.

Inclement Weather, Pandemic, or Substantial Interruption of Instruction

> If in-person classes are suspended, faculty will announce to their students a detailed plan for a change in modality to ensure the continuity of learning. All students must have access to a computer equipped with a web camera, microphone, and Internet access. Resources are available to provide students with these essential tools.

Course Schedule (Tentative)

Details

Week	Date	Topic	Tasks	Things Due
1	August 17 – August 21	1 - Intro to Applied AI	Assigned Reading - Turing Paper.	
2	August 24 – August 28	2 - Ethics	Assigned Reading - Ethics Topic	
3	August 31 – September 4	3 - Modeling Problems		
4	September 7 – September 11	4 - Scaling and Normalization		
5	September 14 – September 18	4 - Distance Metrics and Error Calculation	Classroom Activity 1	
6	September 21 – September 25	5 - RNG		Assignment 2
7	September 28 – October 2	Review	**Mid-term Exam (10/2)**	
8	October 5 – October 9	6 - Local Seach and Opt		
9	October 12 – October 16	7 - ML 1 - Classification vs Clustering	Fall Break	
10	October 19 – October 23	8 - ML 2 - Frameworks and Evaluation	Classroom Activity 2	Assignment 3

11	October 26 – October 30	9 - Neural Networks		
12	November 2 – November 6	10 - CNNs and RNNs		
13	November 9 – November 13	11 - Evolutionary Algoritms	Thanksgiving Break	Assignment 4
14	November 16 – November 20	12 - Gen AI Basics		
15	November 23 – November 27	Finals Review	Extra Credit Courses	Assignment 1,5, Project 1,2

> Disclaimer: This syllabus is tentative and subject to change at the discretion of the instructor. Any changes will be clearly communicated to students in advance.